

RHYTHM GILHOTRA

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OBJECTIVE

Software Engineer with 4+ years of experience developing backend services, microservice architectures, and cloud-native systems. Focused on building reliable, high-availability platforms and improving performance, scalability, and developer efficiency across distributed environments.

TECHNICAL SKILLS

Programming	Python, Java, Kotlin, SQL, Bash
Frameworks	Spring Boot, FastAPI, Flask, gRPC
Cloud	Google Cloud Platform (GCP), Kubernetes, Docker, BigQuery, Pub/Sub
Data	ETL, Kafka, Spark, Hive, Data Modeling
DevOps	GitHub Actions, CI/CD, Prometheus, Grafana, ELF Logging, Terraform, Agile
Architecture	Microservices, API Development, Distributed Systems, System Design

EXPERIENCE

Software Engineer II American Express	May 2023 – Present <i>Gurugram, India</i>
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- Architected and deployed a real-time transaction processing system on GCP using Pub/Sub, Dataflow, and microservices, handling 30M+ events/day and reducing response latency from 15s to under 5s.
- Engineered and integrated A/B infrastructure with telemetry, feature flags, and automated rollback for credit-risk APIs; improved approval accuracy by 15% and reduced false positives by 17%.
- Established CI/CD pipelines with Cloud Build and GitHub Actions, tripling release velocity while maintaining zero Sev-1 incidents and 99.99% system uptime.
- Enhanced system observability using ELF logging, Prometheus, and Grafana, improving incident detection time by 40%.
- Mentored 4+ engineers on modular backend design, distributed tracing, and effective incident response best practices.

Software Engineer I American Express	Aug 2021 – May 2023 <i>Gurugram, India</i>
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- Engineered distributed data processing pipelines using Airflow and BigQuery, optimizing compute efficiency by 75% (6 hrs to 90 mins) and accelerating quarter-close by 2+ days.
- Developed scalable backend services to integrate credit-risk scoring APIs into underwriting systems, improving reliability and reducing early delinquencies by 12% across key markets.
- Designed and deployed real-time monitoring and alerting pipelines with SLA enforcement, retry mechanisms, and failover recovery to ensure 99.9% pipeline uptime.
- Built and automated data aggregation microservices powering Looker dashboards through BigQuery BI Engine, reducing fraud investigation response time by 40%.

Software Engineering Intern American Express	Jan 2021 – Jun 2021 <i>Remote</i>
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- Modernized legacy reporting workflows by introducing a streaming-based architecture for real-time risk dashboards, improving data freshness and reducing report latency by 60%.
- Engineered data processing pipelines in Python and SQL, optimizing query execution and cutting runtime by 50%.

- Built automated data validation and profiling modules, increasing pipeline reliability and eliminating manual QA dependencies.

PROJECTS

Distributed Network Intrusion Detection System (NIDS)

Jan 2020 – Dec 2020

Thapar Institute of Engineering and Technology

- Architected a distributed system to detect malicious network traffic in real time, processing 10K+ packets/sec using Python and TensorFlow.
- Designed modular APIs for data ingestion and rule-based alerting, reducing false positives by 20%.
- Deployed the system on Dockerized containers with live metrics visualization using Grafana.

Scalable Transaction Data Pipeline

Jun 2020 – Dec 2020

American Express (Academic Collaboration)

- Engineered a distributed data pipeline using Hadoop, Spark, and Kafka to process millions of transaction records daily.
- Implemented fault-tolerant ingestion and aggregation layers with SLA monitoring and retries, improving data freshness by 40%.
- Delivered real-time insights and alerting for fraud-risk operations, integrating outputs with downstream analytics systems.

EDUCATION

Bachelor of Engineering in Computer Engineering, Thapar University, Patiala, India

2017 – 2021

Relevant Coursework: Data Structures and Algorithms, Distributed Systems, Operating Systems, Database Systems, Computer Networks, Software Engineering

LEADERSHIP

Mentor & Tech Lead, American Express

2023 – Present

Led backend development initiatives and mentored 4+ engineers on scalable system design, observability, and on-call ownership within the fraud platforms team.

Student Leader, Thapar Institute of Engineering and Technology

2019 – 2021

Organized and delivered technical workshops on backend engineering and data systems; facilitated peer collaboration on distributed project development.

AWARDS

Star Award, American Express

Q1 2025

Recognized for leading real-time fraud detection platform deployments, improving system reliability and scaling performance across global markets.

Analyst of the Quarter, American Express

Q1 2024, Q3 2022

Awarded for delivering high-impact enhancements to risk infrastructure and backend systems, driving measurable operational efficiency.